

Cambridge (CIE) IGCSE Chemistry
Paper 6: Alternative to Practical (Set E)

Saturday 26 April 2025

Afternoon (Time: 1 hour 0 minutes)

Total marks

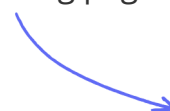
/ 40

Instructions

- Try to complete this mock exam paper in one sitting, under exam conditions. Use all the time available and check your answers to each question at the end before submitting.
- Remember this is PRACTICE. Mistakes are fine and will help you improve in time for the real exam - just do your best.
- You may use a calculator.
- You should show all your working and use appropriate units.

Scan here to mark your mock exam

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1 (a) A student investigated the reaction between sodium thiosulfate and potassium iodate.

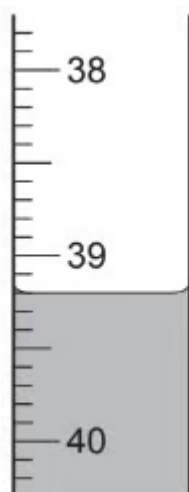
Two experiments were carried out.

Experiment 1

A burette was filled up to the 0.0 cm³ mark with sodium thiosulfate solution. By using a measuring cylinder, 20 cm³ of solution **A** of potassium iodate was placed into a conical flask. Dilute sulfuric acid and potassium iodide were also added to the flask. The flask was shaken to mix the contents and produce a red solution of iodine.

The sodium thiosulfate solution was added to the flask. When the contents of the flask were yellow, 1 cm³ of starch solution was added to the flask. Addition of sodium thiosulfate to the flask was continued until the solution turned colourless.

Use the burette diagram to record the final volume in the table and complete the column in the table of results.



Experiment 2

Experiment 1 was repeated using a different solution of potassium iodate, solution **B**.

Use the burette diagrams to record the volumes and complete the table.

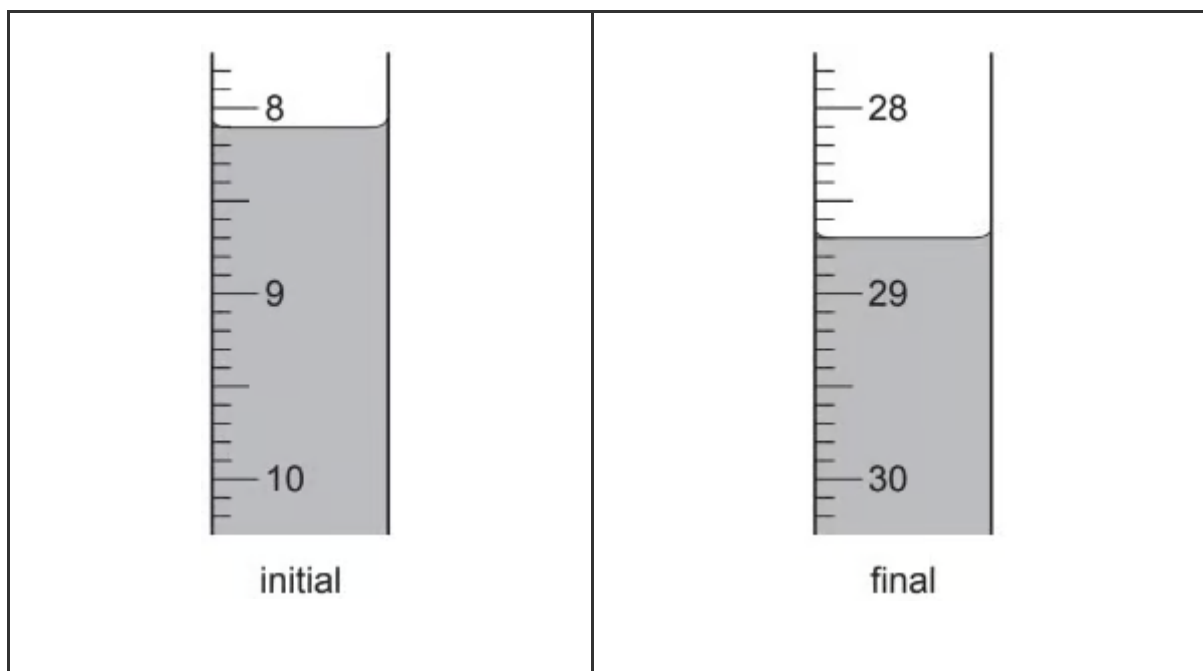


Table of results

Burette readings / cm³

	Experiment 1	Experiment 2
final reading		
initial reading	0.0	8.1
difference		

Suggest why the starch was used.

(5 marks)

(b) i) In which experiment was the greatest volume of sodium thiosulfate solution used?

[1]

ii) Compare the volumes of sodium thiosulfate solution used in Experiments 1 and 2.

[1]

iii) Suggest an explanation for the difference in the volumes.

[2]

(4 marks)

(c) Predict the volume of sodium thiosulfate solution which would be needed to react completely with 10cm^3 of solution **B**.

(2 marks)

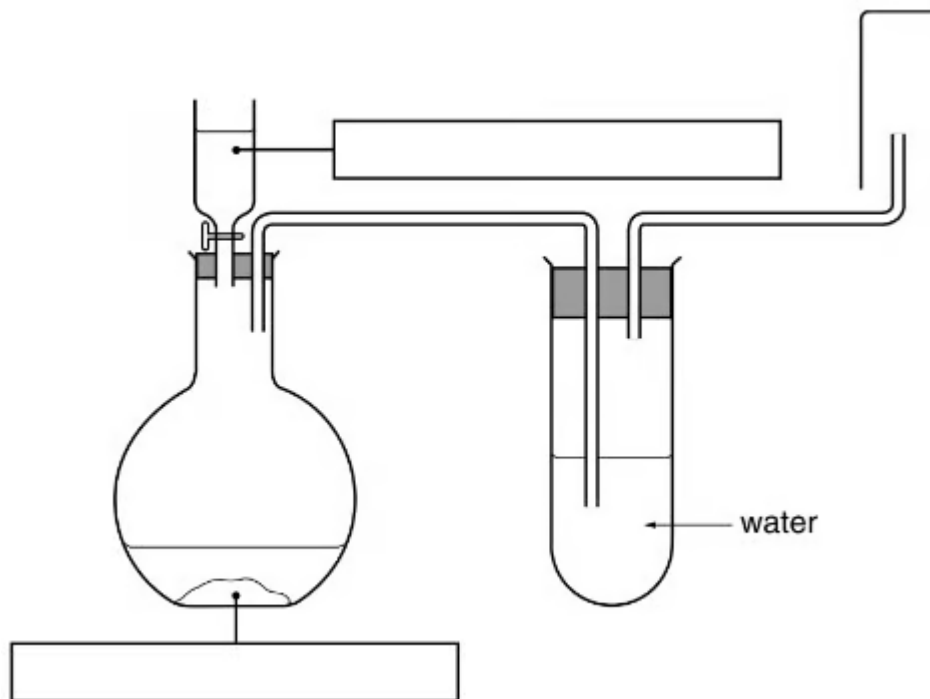
(d) Explain **one** change that could be made to the experimental **method** to obtain more accurate results, without changing the apparatus.

change

explanation

(2 marks)

- 2 (a)** Hydrogen chloride gas is strong-smelling, denser than air and soluble in water. A sample of hydrogen chloride gas can be prepared by adding concentrated sulfuric acid to sodium chloride. Study the diagram of the apparatus used.



Fill in the boxes to show the chemicals used.

.....

.....

(2 marks)

- (b)** Identify and explain **two** mistakes in the diagram.

Mistake 1

[2]

Mistake 2

[2]

.....

.....

(4 marks)

(c) State **one** precaution that should be taken when carrying out this experiment.

(1 mark)

- 3 (a)** A student investigates the reaction between dilute hydrochloric acid and two different aqueous solutions of sodium hydroxide labelled solution **A** and solution **B**.

The student does two experiments.

Experiment 1

The student:

- rinses a burette with dilute hydrochloric acid
- fills the burette with dilute hydrochloric acid
- runs out some of the acid so that the level of acid is on the burette scale
- uses a measuring cylinder to pour 25 cm³ of solution **A** into a conical flask
- adds five drops of thymolphthalein indicator to the conical flask
- swirls the flask while adding the acid from the burette to the conical flask until the solution just changes colour.

Experiment 2

The student:

- empties and rinses the conical flask with distilled water
- repeats Experiment 1 using solution **B** instead of solution **A**.

Use Fig. 2.1 and Fig. 2.2 to record the readings for Experiments 1 and 2 in Table 2.1 and complete Table 2.1.

Experiment 1

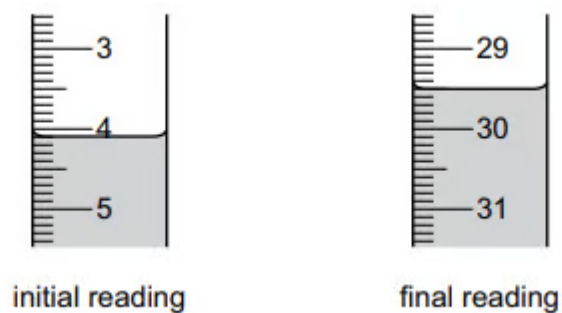


Fig. 2.1

Experiment 2

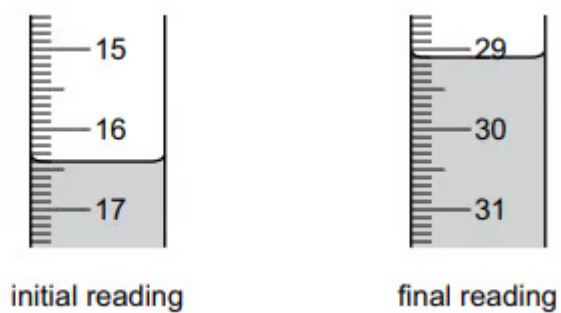


Fig. 2.2

Table 2.1

	Experiment 1	Experiment 2
final burette reading / cm ³		
initial burette reading / cm ³		
volume of dilute hydrochloric acid added / cm ³		

.....
.....
.....
.....
(4 marks)

- (b) (i) State which solution of sodium hydroxide, solution **A** or solution **B**, is the more concentrated.

Explain your answer.

[1]

- (ii) Deduce the simplest whole number ratio of concentration of solution **A** : concentration of solution **B**.

[1]

.....
.....
(2 marks)

- (c) State the volume of hydrochloric acid needed if Experiment 1 is repeated using 10 cm³ of solution **A**.

.....
.....
(2 marks)

(d) In Experiment 2 the conical flask is rinsed with distilled water.

(i) Suggest why the conical flask is rinsed with distilled water.

[1]

(ii) The conical flask is **not** dried after it is rinsed with distilled water.

Suggest why the conical flask is **not** dried.

[1]

(2 marks)

(e) State the effect, if any, on the volume of dilute hydrochloric acid used in Experiment 1 if the solution of sodium hydroxide is warmed before adding the dilute hydrochloric acid.

Give a reason for your answer.

effect on volume

reason

(2 marks)

- 4 (a)** Two metal salt solutions, **X** and **Y**, were analysed. Solution **X** was iron(II) chloride. The tests on **X** and **Y**, and some of the observations, are given in the following tables. Complete the observations in the tables.

test	observations
tests on solution X	
Appearance of solution X	

.....
(1 mark)

- (b)** The solution was divided into four equal portions.

tests	observations
Dilute nitric acid and aqueous silver nitrate were added to the first portion of solution	

.....
(1 mark)

(c)

tests	observations
Aqueous sodium hydroxide was added to the second portion of solution and the mixture shaken	

(2 marks)

(d)

tests	observations
Excess aqueous ammonia was added to the third portion of solution.	

(1 mark)

(e)

tests	observations
An oxidising agent was added to the fourth portion of the solution. Aqueous sodium hydroxide was then added to the mixture.	

(2 marks)

(f)

tests	observations
<u>tests on solution Y</u> The solution was divided into three equal portions. Dilute hydrochloric acid was added to the first portion of the solution.	white precipitate formed
Aqueous sodium hydroxide was added to the second portion of the solution and the mixture shaken. Aluminium powder was added to the mixture and it was warmed gently. The gas given off was tested with damp red litmus paper.	effervescence pungent gas evolved, litmus paper turned blue
Aqueous potassium iodide was added to the third portion of the solution.	pale yellow precipitate

What conclusions can you draw about solution Y?

(1 mark)